Protocols – SpyTag/SpyCatcher

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

*This protocol designs the **genetic constructs needed to join two enzymes into a single complex** through the SpyTag/SpyCatcher system, which forms a spontaneous, irreversible isopeptide bond when the two partners meet, with no enzyme, cofactor or chemical crosslinker required. One enzyme carries the short SpyTag peptide and the other the larger SpyCatcher domain, so the complex assembles by itself when both purified proteins are mixed.*

Materials list

Input data

- Amino-acid and nucleotide sequences of both enzymes.
- Sequences of **SpyTag** and **SpyCatcher** (SpyTag/SpyCatcher 003).
- Structures of both enzymes, experimental or predicted.

Software

- Structure prediction tool (AlphaFold or Boltz-2)
- **PyMOL**, to superpose the models and inspect the active sites
- Sequence editor able to translate and check reading frames (ExPASy Translate)

Steps

A. Assigning the elements:

1. Decide which enzyme carries the SpyTag and which carries the SpyCatcher. As a starting criterion, assign the **smaller element to the larger protein**: adding the bulkier domain to a large or oligomeric enzyme increases the risk of interfering with folding, with oligomerisation or with access to the active site.
2. Treat this as a hypothesis, not a conclusion. Record the reciprocal assignment as a construct to be designed and tested as well, since size alone does not predict how a fusion will behave.

B. Choosing where to attach the fusion:

1. Model each enzyme **with its complete fusion**, so that the tag or catcher is included in the prediction rather than added afterwards on paper.
2. Repeat the modelling for each attachment point under consideration, at the **N-terminus** and at the **C-terminus** of each enzyme.
3. Superpose the models of the two partners in PyMOL, in the orientation the assembled complex would adopt.
4. Inspect each combination against two criteria: the fusion must sit **away from the catalytic regions**, and it must leave the **two active sites facing each other**, since the proximity of the two sites is what substrate channelling depends on.
5. Choose the combination that best satisfies both criteria. Keep images of the models and of the superposition as a record of the decision.

C. Building the construct sequence:

1. Assemble the fusion sequence in the chosen order: tag or catcher, linker (if used; we used: GSSGS), and the enzyme.
2. Add the restriction sites required for cloning at each end (NdeI, NcoI, BamHI, SalI, XhoI as appropriate), and insert any additional nucleotides needed to keep the reading frame after digestion.
3. Translate the complete nucleotide sequence and check that the protein comes out continuous and in frame, with no premature stop codon and no unintended internal restriction sites.
4. Check the ends: an initiation codon at the start, and a stop codon at the end only if you do not want the vector's C-terminal tag to be translated.
5. Order the construct as a synthetic gene, codon-optimised for the expression host.

D. Experimental plan to follow (not covered by this protocol):

1. Express and purify each fusion **on its own**, and compare its expression, solubility and activity with the corresponding unfused enzyme, which is the control.
2. Combine only the fusions that retain their activity, and compare the assembled complex with the **free enzyme mixture** in the coupled assay. That comparison is what shows whether joining the enzymes actually helps.
3. Where possible, compare several attachment combinations, since the position of the fusion may be what limits the system.

Resources & Downloads

- Keeble & Howarth (2020), *Chemical Science*: https://doi.org/10.1039/d0sc01878c

Troubleshooting

- **The reading frame breaks after digestion.** Count the nucleotides contributed by each restriction site and add the extra bases needed to restore the frame. Always verify by translating the full sequence, not by inspecting it.

- **The modelled complex leaves the two active sites far apart.** Try the other terminus, or the reciprocal assignment of tag and catcher. This is exactly why several combinations should be designed rather than only one.

- **The fusion is predicted to sit close to the active site.** Choose the opposite terminus, or add a flexible linker to move the element away, keeping in mind that a longer linker also increases the distance between the two active sites.

- **We could not test the designs.** Our constructs were designed in full but not taken to the bench within the time available, and only one of the two possible assignments was designed. Build in enough time for expression and assay, since the design itself proves nothing on its own.